Quantification: 699_HYP_HP_z25c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 5 |
| interior lacuna count | 5 |
| frame-edge lacuna count | 0 |

Interior lacunae only: mean, median and sample SD (n = 5)
---------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1265.6 | 1302 | 204.353 | px^2 |
| major_axis_length_px | lacuna shape | 65.1494 | 67.6703 | 15.5372 | px |
| minor_axis_length_px | lacuna shape | 28.7664 | 30.4825 | 3.3967 | px |
| aspect_ratio | lacuna shape | 2.3204 | 2.348 | 0.7541 | unitless |
| eccentricity | lacuna shape | 0.8701 | 0.9048 | 0.0835 | unitless |
| solidity | lacuna shape | 0.815 | 0.7724 | 0.0791 | unitless |
| roots_count | headline | 5.4 | 6 | 4.3359 | count |
| ring_length_r30_px | headline | 258.6 | 274 | 144.431 | px |
| ring_length_r60_px | reported | 601 | 583 | 327.08 | px |
| owned_length_px | ownership-dependent | 598.403 | 318.38 | 763.337 | px |
| edge_count | ownership-dependent | 34.2 | 14 | 45.0688 | count |
| mean_edge_length_px | ownership-dependent | 14.7823 | 17.1228 | 8.6293 | px |
| perimeter_px | normalised | 211.097 | 169.924 | 78.5409 | px |
| ring_area_r30_px2 | normalised | 6763.4 | 6994 | 1633.61 | px^2 |
| ring_area_r60_px2 | normalised | 17800 | 19484 | 4879.3 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 1 | 1 | 0 | unitless |
| ring_density_r30 | normalised | 0.0357005 | 0.0391764 | 0.0135787 | px^-1 |
| ring_density_r60 | normalised | 0.0321157 | 0.0281602 | 0.0101908 | px^-1 |
| roots_per_100px_perimeter | normalised | 2.4402 | 2.5927 | 2.0693 | per 100 px |
| ring_attached_length_r30_px | network v2 | 211 | 194 | 158.961 | px |
| ring_attached_length_r60_px | network v2 | 431.6 | 341 | 338.718 | px |
| ring_length_w_r30_px | network v2 | 290.297 | 305.551 | 162.502 | px |
| ring_length_w_r60_px | network v2 | 670.724 | 637.931 | 370.905 | px |
| sholl_crossings_r10 | network v2 | 6.4 | 6 | 4.5056 | count |
| sholl_crossings_r20 | network v2 | 9.6 | 9 | 5.8992 | count |
| sholl_crossings_r30 | network v2 | 7 | 7 | 2.5495 | count |
| loop_count | loops, report only | 1.8 | 0 | 4.0249 | count |
| loop_edge_share | loops, report only | 0.0625 | 0 | 0.1398 | share |
| ring_width_mean_r30_px | width | 6.5562 | 6.1521 | 0.6917 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.04225e+06 | px^2 |
| total_skeleton_length_px | 17504 | px |
| canalicular_length_density_per_px | 0.0167945 | px^-1 |
| junction_count | 333 | count |
| junction_density_per_px2 | 0.0003195 | px^-2 |
| lacuna_count | 5 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 4.8e-06 | px^-2 |
| skeleton_component_count | 334 | count |
| mean_component_length_px | 52.4072 | px |
| median_component_length_px | 15 | px |
| field_length_density_w_per_px | 0.018758 | px^-1 |
| field_density_without_flagged_per_px | 0.0150603 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 785 | px |
| unattached_component_count | 39 | count |
| canalicular_length_density_attached_per_px | 0.0167945 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0175477 | px^-1 |
| loop_count | 30 | count |
| loop_edge_share | 0.1234 | share |
| edge_count | 989 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.8823 | px |
| width_median_px | 6.3246 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 12347 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1197 | 51.8325 | 30.7893 | 1.6835 | 0.8045 | 0.9089 | -1.2998 | 286.212 | 636.439 | False |
| 2 | 1418 | 83.5326 | 24.5778 | 3.3987 | 0.9557 | 0.7698 | -1.1714 | 319.255 | 630.773 | False |
| 3 | 1461 | 75.7643 | 32.2675 | 2.348 | 0.9048 | 0.7334 | -1.4732 | 556.736 | 495.522 | False |
| 4 | 1302 | 67.6703 | 25.7149 | 2.6316 | 0.925 | 0.8906 | 1.4126 | 532.052 | 690.266 | False |
| 5 | 950 | 46.9475 | 30.4825 | 1.5401 | 0.7605 | 0.7724 | -1.05 | 698.178 | 753.763 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 74 | 242 | 0 | 0 | 0 | 6.069 |
| 2 | 6 | 183 | 427 | 210.095 | 12 | 17.508 | 6.055 |
| 3 | 10 | 463 | 1118 | 1917.75 | 112 | 17.1228 | 7.6346 |
| 4 | 2 | 299 | 583 | 318.38 | 14 | 22.7414 | 6.8704 |
| 5 | 9 | 274 | 635 | 545.789 | 33 | 16.5391 | 6.1521 |

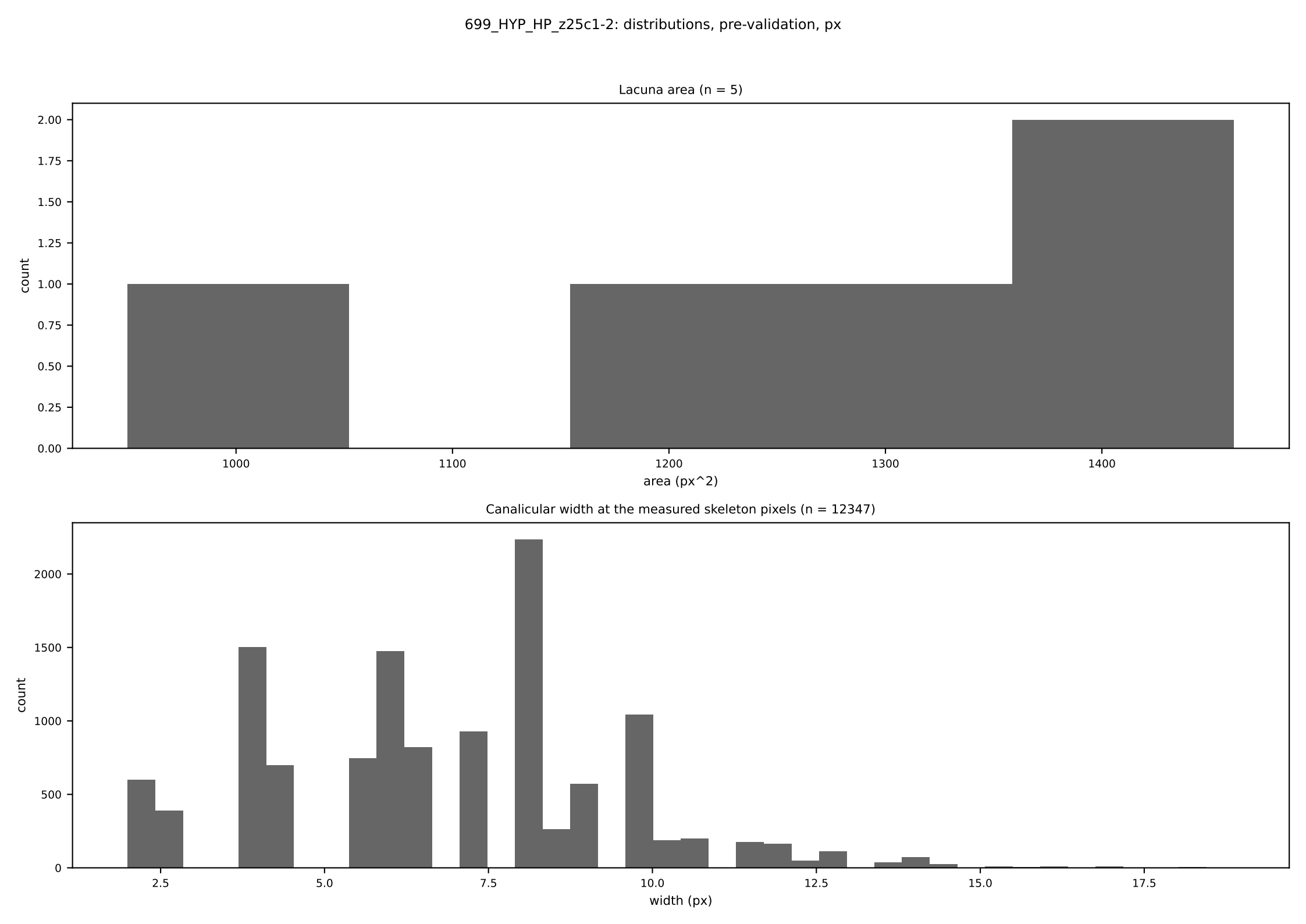
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 150.167 | 4208 | 10231 | 1 | 1 | 0.0175855 | 0.0236536 | 0 |
| 2 | 231.421 | 6473 | 15957 | 1 | 1 | 0.0282713 | 0.0267594 | 2.5927 |
| 3 | 340.049 | 8613 | 22625 | 1 | 1 | 0.0537559 | 0.0494144 | 2.9408 |
| 4 | 169.924 | 7529 | 20703 | 1 | 1 | 0.0397131 | 0.0281602 | 1.177 |

5 163.924 6994 19484 1 1 0.0391764 0.0325908 5.4904

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 24 | 77 | 82.1838 | 270.137 | 1 | 3 | 4 |
| 2 | 143 | 271 | 204.681 | 471.191 | 6 | 5 | 6 |
| 3 | 457 | 975 | 518.919 | 1259.79 | 13 | 14 | 11 |
| 4 | 194 | 341 | 340.149 | 637.931 | 4 | 17 | 7 |
| 5 | 237 | 494 | 305.551 | 714.571 | 8 | 9 | 7 |


699_HYP_HP_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 5)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 12347)
count
width (px)